Learning Money by Living It

A pixel-art life simulation that teaches financial literacy by letting students live it

Team No.: 97

M2 — Progress report

Status at 03:30, 11 Oct: Stage 1 of the game is playable end to end on a phone-sized browser; the game engine is unit-tested; the content pack validates against its schema; the report service emails a PDF to a sandbox inbox. Code is on `main`.

1. Current Approach and Changes Since M1

Unchanged: the three-stage life simulation, money-battle screen, teach-after-the-choice loop with abstract animations, mistake bank, persona framing, difficulty buckets, before/after quiz, emailed report, NGO join codes, guardrail.

Decided since M1 (M1 left these open):

- **Economy.** Everyone starts with $S = ₹5{,}000$ in-game per month in Stage 1; in-game price $=$ real price $\times S/$ monthly pocket money (band midpoints ₹200, ₹800, ₹2,000, ₹4,000 from the indirect weekly-spend question). Stages 2–3 use fixed reference incomes (₹20,000 and ₹50,000 real $\rightarrow$ ₹25,000 and ₹60,000 in-game).
- **Time.** Ten monthly chapters: 4 (Intermediate) $+$ 3 (first salary) $+$ 3 (earning more), then the epilogue at 60. Stage jumps skip 5, 6 and 25 years so compounding is visible.
- **Happiness.** 0–100, starts at 70, decays 3 per chapter; choices move it -30 to $+15$; zero restarts the task; below 30 locks mini-games.
- **Mini-games.** Two, skippable: *Kirana Rush* (sort needs from wants) and *Scam or Legit* (swipe fake messages). A win pays 8% of monthly income as a "side gig" and $+10$ happiness.
- **Hints and guide.** A guide character delivers lesson lines; hints switch on after two failures of the same task.
- **Second language.** Telugu for onboarding, HUD and two tasks; a Hindi file proves "add a language $=$ add a file".
- **Stack.** TypeScript + Phaser 3 + Vite (PWA, offline), Vitest, zod schema validation, a small Node report service (nodemailer $\rightarrow$ sandbox inbox), static NGO dashboard mock.

2. Experiments Run So Far and Results

- **Economy scaler check.** ₹200 cricket bat $\rightarrow$ ₹200 / ₹400 / ₹1,000 in-game for students with ₹5,000 / ₹2,500 / ₹1,000 monthly pocket money (same share of their money as in real life). Unit-tested.
- **Seeded simulations.** Three policies (*always good*, *random*, *always worst*) over 10 chapters on the Stage 1 content, same seed: **[TABLE: wallet/happiness at chapter 4, detours, restarts per policy]**. The "always worst" run restarts on the scam task and ends broke; "always good" ends solvent with the mistake bank empty.
- **Engine unit tests. [N]** tests, all passing: RNG reproducibility, scaler, delayed/repeating effects, task selection (pillar rotation, invest gate, returning mistakes), restart/detour routing, save/resume round-trip.
- **Content validation. [N]** main tasks, **[N]** detours, **[N]** persona framings, **[N]** explainer animations, **[N]** random events; every cross-reference (concept, explainer, detour, place, event) resolves; double-blind string check passes.
- **Play test. [N]** peers played Stage 1 unguided on a phone: time to first battle **[s]**, help requests **[n]**, **[notes]**.
- **Device/network.** Production build on Chrome "Slow 3G" + 4$\times$ CPU throttle: first battle in **[s]**; full offline reload works via the service worker.
- **Report pipeline.** A seeded game produced a PDF report that was emailed to a sandbox inbox with the guardian in CC; a no-network fallback stores the message locally.

3. Progress on Testing

Learning gain	Before/after quiz implemented end to end; NCFE-aligned scoring (8 knowledge, 9 behaviour, 3 attitude) and safety score computed from tagged choices	**[status]**
Usability	Unguided peer play tests of Stage 1	**[status]**
Engine correctness	Unit tests + seeded simulations + save/resume round-trip	**[status]**
Devices & network	Throttled-Chrome timing, offline play-through	**[status]**
Language	Validator checks every Telugu key exists in English; Telugu onboarding screenshots	**[status]**
Report & privacy	Sandbox inbox send; nickname-only state; no OTP/PIN/Aadhaar ever requested	**[status]**

4. Plan for the Remaining Time

- **04:00–08:30.** Stages 2–3 content; investing screen with four made-up assets and the epilogue projection; "what if" replay of the biggest mistake; after-quiz and end report; both mini-games; Telugu onboarding; NGO dashboard mock; throttled-device numbers.
- **08:30 feature freeze.** Cross play tests, bug fixes, demo video, final report, README with asset credits.
- **Cut list, in order:** second mini-game; Telugu beyond onboarding; Stage 3 depth; town map (chapter map goes

straight to the battle); live email (download only); what-if screen (text comparison).